HACKTIVATE

24-Hour Predictive Systems Hackathon

IEEE Student Branch, RUAS Peenya Campus

Event-Specific Rules & Regulations

1. Eligibility and Team Composition

1. The event is open to students from any college or university, across all branches and all years of study. Participants from outside RUAS are welcome.
2. Teams must consist of a minimum of 2 and a maximum of 4 members. Individual entries are not permitted.
3. A participant may be a member of only one team. Members cannot be exchanged between teams once registration closes.
4. Each team must nominate one member as team lead, who serves as the point of contact for all organiser communication during the event.
5. All team members must be present at the venue for the full duration of the event.

2. Theme and Problem Selection

1. All builds must address the event theme, Predictive Systems: a working hardware and software system that senses real-world data and predicts an outcome or trend, such as failure, demand, anomaly, or an early-warning condition.
2. Teams define their own application within the theme. Organisers do not prescribe the use case, technology stack, or architecture.
3. Each team confirms its chosen application with the organisers at the end of the ideation phase. It may not be changed after the mid-hackathon checkpoint.

3. Build Rules

1. The core of a team's submission must be designed and built live during the event. A team may not enter an existing project, or a modified version of one, as its primary work.
2. Existing work a team has previously built may be used as a supporting component — a library, module, utility, or boilerplate the team already owns. It must serve the new build, not constitute it. Teams must declare any such component to the judges at demonstration.
3. Designs must be produced using KiCad or an organiser-approved equivalent, with simulation verification of the sensing and hardware layer before assembly begins.
4. Open-source libraries, public APIs, publicly available datasets, and approved tooling may be used freely. Use of AI coding assistants is permitted; teams should be prepared to explain any part of their submission when asked.
5. Teams must bring their own laptops with development environments pre-installed. Organisers provide power, connectivity, and workspace.
6. Any design, code, or submission plagiarised from another team results in immediate disqualification of the entire team.

4. Phase Checkpoints

The event runs in defined phases: ideation, design and simulation, build and sensor integration, predictive-layer integration, and final demonstration.

- Each phase carries a guideline checkpoint. Teams significantly behind schedule may receive mentor support but forfeit time-bonus marks.

- At the mid-hackathon checkpoint, every team gives a brief progress update to mentors and receives guidance for the overnight session.

5. Overnight Conduct and Safety

1. Participants must remain within the designated venue for the full 24 hours. Sign-out is permitted only with organiser approval.
2. A supervised rest area is provided. Participants are strongly encouraged to rest in shifts overnight for their own safety and wellbeing.
3. Soldering and assembly work must be carried out only at designated stations, observing all safety instructions given by organisers.
4. Organiser supervision and security are maintained throughout the night. Participants must follow instructions from organisers and security staff at all times.
5. First-aid provision is available for the full duration. Any injury or incident must be reported to organisers immediately.
6. Damage to venue property or equipment is the responsibility of the participants concerned.

6. Submission and Demonstration

1. Judging is based on a live, powered-on demonstration presented to the judging panel. Recorded demonstrations and slide-only submissions are not accepted.
2. Non-functional builds are scored on design and build quality only, and forfeit marks under the working-demonstration criterion.
3. Each team is allotted a fixed demonstration slot followed by questions from the judges. Teams must be ready at their allotted time.
4. Teams must submit their schematics, source code, and bill of materials to the organisers before demonstrations begin.

7. Judging Criteria

Criterion	Marks
Concept — the problem addressed and the soundness of the approach taken	20
Design correctness — schematic passes Design Rule Check with zero errors	20
Routing and layout efficiency — trace length, via count, and layer usage minimised	20
Build quality — component placement, assembly, and sensor integration	20
Working demonstration — the predictive system powers on and performs as intended	20
Total	**100**

Judges' and organisers' decisions are final and binding in all matters.

8. General

1. Participants are expected to conduct themselves respectfully towards fellow participants, volunteers, mentors, judges, and organisers at all times.
2. Participants must comply with all applicable University and TechFest regulations throughout the event.
3. Organisers reserve the right to disqualify any team for misconduct, plagiarism, unsafe practice, or breach of these rules.

Organised by the IEEE Student Branch, RUAS Peenya Campus.